

Ngoc Minh Quang Nguyen

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EDUCATION

Ph.D. Statistics, Carnegie Mellon University 2022 – present
Thesis: Statistical modeling problems for player movement in sports with tracking data
Committee: Ron Yurko (chair), Will Townes, Chad Schafer, Sam Ventura, Jun Yan

M.S. Statistics, Carnegie Mellon University 2022 – 2024

M.S. Applied Statistics, Loyola University Chicago 2020 – 2021

B.S. Mathematics & Data Science, Wittenberg University 2017 – 2020
Summa Cum Laude, Departmental Honors, Phi Beta Kappa

PUBLICATIONS

JOURNAL ARTICLES

R. Yurko, **Q. Nguyen**, K. Pelechris. NFL Ghosts: A framework for evaluating defender positioning with conditional density estimation. 2026. *Annals of Applied Statistics* **20**(1), 873–892. [doi:10.1214/25-AOAS2132](https://doi.org/10.1214/25-AOAS2132); [arXiv:2406.17220](https://arxiv.org/abs/2406.17220).

Q. Nguyen, R. Yurko. A multilevel model with heterogeneous variances for snap timing in the National Football League. 2025. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*. [doi:10.1093/jrssi/qnaf185](https://doi.org/10.1093/jrssi/qnaf185); [arXiv:2502.16313](https://arxiv.org/abs/2502.16313).

Q. Nguyen, R. Jiang, M. Ellingwood, R. Yurko. Fractional Tackles: Leveraging Player Tracking Data for Within-Play Tackling Evaluation in American Football. *Scientific Reports (Nature Portfolio)* **15**, 2148. 2025. [doi:10.1038/s41598-025-85993-1](https://doi.org/10.1038/s41598-025-85993-1); [arXiv:2403.14769](https://arxiv.org/abs/2403.14769).

Q. Nguyen, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *The American Statistician* **78**(2), 199–208. 2024. [doi:10.1080/00031305.2023.2242442](https://doi.org/10.1080/00031305.2023.2242442); [arXiv:2305.10262](https://arxiv.org/abs/2305.10262).

Editors' Choice Collection.

Q. Nguyen, G. J. Matthews. Filling the Gaps: A Multiple Imputation Approach to Estimating Aging Curves in Baseball. *Journal of Sports Analytics* **10**(1), 77–85. 2024. [doi:10.3233/JSA-240744](https://doi.org/10.3233/JSA-240744); [arXiv:2210.02383](https://arxiv.org/abs/2210.02383).

N. M. Lemon, L. K. Taylor, M. A. Rech, **Q. Nguyen**, G. J. Matthews, G. Lew, S. Lovett. Utility of D-dimer in predicting pulmonary embolism in patients with COVID-19 presenting to the emergency department. *Journal of the American College of Emergency Physicians Open* **5**(4), e13237. 2024. [doi:10.1002/emp2.13237](https://doi.org/10.1002/emp2.13237).

B. S. Baumer, G. J. Matthews, **Q. Nguyen**. Big Ideas in Sports Analytics and Statistical Tools for their Investigation. *Wiley Interdisciplinary Reviews: Computational Statistics* **15**(6), e1612. 2023. [doi:10.1002/wics.1612](https://doi.org/10.1002/wics.1612); [arXiv:2301.04001](https://arxiv.org/abs/2301.04001).

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *New England Journal of Statistics in Data Science* **1**(1), 95–101. 2023. [doi:10.51387/21-NEJSDS1](https://doi.org/10.51387/21-NEJSDS1); [arXiv:2105.09881](https://arxiv.org/abs/2105.09881).

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Journal of Data Science* **20**(2), 156–167. 2022. [doi:10.6339/22-JDS1042](https://doi.org/10.6339/22-JDS1042); [arXiv:2111.05310](https://arxiv.org/abs/2111.05310).

SUBMITTED & WORKING PAPERS

Q. Nguyen, R. Yurko. Bayesian multilevel step-and-turn models for evaluating player movement in American football. 2026+. [arXiv:2603.17866](#).

Q. Nguyen, R. Yurko. A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. 2025+. [arXiv:2507.06122](#).

Q. Nguyen, P. Campana, N. M. D. Niezink. Co-evolution of cooperation and conflict among organized crime groups: Evidence from the United Kingdom. 2025+.

R. Yurko, R. Jiang, M. Ellingwood, **Q. Nguyen**. Momentum Interference: A Hierarchical Framework for Modeling Tackling Ability in American Football. 2024+.

TEACHING EXPERIENCE

AT CARNEGIE MELLON UNIVERSITY

36-315: Statistical Graphics and Visualization Summer 2026
Undergraduate course on creating, critiquing, and presenting data visualizations from a statistical perspective. <http://36-315-summer26.netlify.app>

Summer Undergraduate Research Experience in Statistics Summer 2024 & 2025
Two-month summer initiative with 35–40 students selected each year nationally with an emphasis on diversity. Program offers two themes: [Carnegie Mellon Sports Analytics Camp](#) and [UnitedHealth Group Bridges to Healthcare Technology](#). Student experience includes hands-on training on statistics and data science, culminating in a client-facing capstone with real-world problems and datasets.
Responsibilities: coordinated program curriculum and schedule, developed lecture and lab materials, taught daily lectures, and advised student projects. <https://36-sure.github.io>

CONFERENCE WORKSHOPS

Player Tracking Data Workshop, Connecticut Sports Analytics Symposium April 2025 & 2026
Advanced-level workshops on building metrics (2025; <http://getstrained.github.io/talks/csas>) and unsupervised learning (2026; <https://qntkhvn.github.io/timing/csas>) with tracking data. Developed materials and delivered live instruction with hands-on code demo.

Big Data Bowl Workshop, Carnegie Mellon Sports Analytics Conference November 2023
Workshop sponsored by SumerSports on tracking data and NFL Big Data Bowl. Developed materials and delivered live instruction with hands-on code demo. <http://getstrained.github.io/talks/cmsac>

AT LOYOLA UNIVERSITY CHICAGO

STAT 103: Fundamentals of Statistics Spring 2022
Undergraduate course on introductory statistics. Taught 3 sections.

OTHER ACADEMIC EXPERIENCE

GRADUATE TEACHING ASSISTANT

36-460/660: Sports Analytics, Carnegie Mellon University Spring 2024, 2025 & 2026
36-468/668: Text Analysis, Carnegie Mellon University Fall 2023, 2024 & 2025
Summer Undergraduate Research in Statistics, Carnegie Mellon University Summer 2023

36-402/608: Advanced Methods for Data Analysis, Carnegie Mellon University
36-401/607: Modern Regression, Carnegie Mellon University

Spring 2023
Fall 2022

ACADEMIC SUPPORT

Math Workshop Head Tutor, Wittenberg University 2018 – 2020
Faculty Aide, Wittenberg University 2018 – 2020
S.U.M. (Student Understanding Math) Assistant, Clarkston Senior High School 2017

PRESENTATIONS

TALKS

Q. Nguyen, R. Yurko. A Bayesian circular mixed-effects model for evaluating change of direction in American football. *Joint Statistical Meetings*. Boston, MA. August 2026. (Contributed)

Q. Nguyen, R. Yurko. NFL CoD: A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2025. (Invited)

Q. Nguyen. Advances in pass rush and change of direction evaluation in American football with player tracking data. *U.S. Olympic and Paralympic Performance Innovation Summit*. Colorado Springs, CO. August 2025. (Invited)

Q. Nguyen, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. *Joint Statistical Meetings*. Nashville, TN. August 2025. (Contributed)

Q. Nguyen, R. Yurko. Step-and-turn modeling of individual player movement in American football. *Conference on Predictive Inference in Sports*. Ames, IA. June 2025. (Invited)

Q. Nguyen, R. Yurko. Explaining variability in player turning behavior in the National Football League. *CMSAC Football Analytics Workshop*. Pittsburgh, PA. May 2025. (Invited)

Q. Nguyen. Advances in pass rush and tackling evaluation in American football with player tracking data. *2024 Joint Statistical Meetings*. Portland, OR. August 2024. (Invited)

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Fractional tackles. *UConn Sports Analytics Symposium*. Storrs, CT. April 2024. (Invited)

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. *NFL Big Data Bowl 2024*. Indianapolis, IN. February 2024. (Invited)

Finalist (Top 5 + \$12,500 prize + Presented at 2024 NFL Scouting Combine)

Q. Nguyen. Opportunities in sports analytics with player tracking data: A case study of NFL pass rush evaluation. *Data Science Seminar, Loyola University Chicago*. Chicago, IL. November 2023. (Invited)

Q. Nguyen, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2023. (Invited, featured talk)

G. J. Matthews, **Q. Nguyen**. STRAIN: Sacks, Tackles, Rushing, Aggression INdex. *NFL Big Data Bowl 2023*. Indianapolis, IN. March 2023. (Invited)

Finalist (Top 8 + \$10,000 prize + Presented at 2023 NFL Scouting Combine)

Q. Nguyen, G. J. Matthews. Estimating Aging Curves: Using Multiple Imputation to Examine Career Trajectories of MLB Offensive Players. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. October

2022. (Invited)

Finalist, Open Methodology Track, Reproducible Research Competition

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Cascadia Symposium on Statistics in Sports*. Vancouver, BC. September 2022. (Invited)

Q. Nguyen, H. Butler, G. J. Matthews. An Examination of Sport Climbing's Competition Format and Scoring System. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2021. (Invited, virtual)

Winner, Open Methodology Track, Reproducible Research Competition

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *Loyola Graduate School Interdisciplinary Research Symposium*. Chicago, IL. March 2021. (Contributed, virtual)

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *Midwest Sports Analytics Meeting*. Pella, IA. November 2020. (Contributed, virtual)

POSTERS

Q. Nguyen, H. Butler, G. J. Matthews. Concerns Regarding Sport Climbing Competition Format and Scoring System: Who's Suffering? *UConn Sports Analytics Symposium*. October 2021. (Virtual)

Winner, Student Poster Award

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *NextGen: Data Science Day, New England Statistical Society*. November 2020. (Virtual)

Co-Winner, Poster Competition, Undergrad/MS Group

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *Carnegie Mellon Sports Analytics Conference*. October 2020. (Virtual)

Q. Nguyen. Poisson Modeling and Predicting English Premier League Goal Scoring. *UConn Sports Analytics Symposium*. October 2020. (Virtual)

TUTORIALS & WORKSHOPS

Q. Nguyen. Unsupervised learning with tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. Storrs, CT. April 2026.

Q. Nguyen. Rcpp: A primer. *StatBytes Statistical Computing Seminars, Carnegie Mellon Statistics & Data Science*. Pittsburgh, PA. November 2025.

Q. Nguyen. Building a performance metric with player tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. New Haven, CT. April 2025.

Q. Nguyen. Get Trained & Get Strained: The Ingredients of a Big Data Bowl Metric. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2023.

GUEST LECTURES

Q. Nguyen. Player tracking data: space, time, interactions & heterogeneity. *Summer Undergraduate Research Experience in Statistics: Carnegie Mellon Sports Analytics Camp*. Pittsburgh, PA. July 2026.

Q. Nguyen. Bayesian multilevel modeling in the wild: Beyond homogeneity. *36-460/660: Sports Analytics, Carnegie Mellon University*. Pittsburgh, PA. March 2025 & 2026.

Q. Nguyen. Web Scraping: A Primer. *DSCI 401: Introduction to Data Science, Loyola University Chicago*. Chicago, IL. November 2023.

Q. Nguyen. Big Data Bowl: Overview, Getting Started, My Experience, and more. *Summer Undergraduate Research Experience in Statistics: Carnegie Mellon Sports Analytics Camp*. Pittsburgh, PA. July 2023.

Q. Nguyen. Web Scraping: A Primer. *Summer Undergraduate Research Experience in Statistics: Carnegie Mellon Sports Analytics Camp*. Pittsburgh, PA. July 2023.

Q. Nguyen. Introduction to NumPy. *COMP 231: Data Structures & Algorithms for Informatics, Loyola University Chicago*. Chicago, IL. March 2022.

Q. Nguyen. Probability Distributions: Normal, Exponential, and Gamma. *STAT 203: Introduction to Probability & Statistics, Loyola University Chicago*. Chicago, IL. March 2022.

MISCELLANEOUS ARTICLES

Q. Nguyen, R. Yurko (joint work with NFL). Introducing the NFL Draft Tier Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/introducing-the-nfl-draft-tier-chart>

Q. Nguyen, R. Yurko (joint work with NFL). Building an NFL Draft Value Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/building-an-nfl-draft-value-chart>

Q. Nguyen, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. NFL Big Data Bowl 2025. <https://www.kaggle.com/tindata/down-set-hut>

Q. Nguyen, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. NFL Big Data Bowl 2024. <https://www.kaggle.com/tindata/momentum-based-fractional-tackles>

G. J. Matthews, **Q. Nguyen.** STRAIN: Sacks, Tackles, Rushing, Aggression INDEX. NFL Big Data Bowl 2023. <https://www.kaggle.com/statsinthewild/strain-sacks-tackles-rushing-aggression-index>

B. S. Baumer, **Q. Nguyen,** G. J. Matthews. CRAN Task View: Sports Analytics. The Comprehensive R Archive Network. 2022. <https://cran.r-project.org/web/views/SportsAnalytics.html>

HONORS & AWARDS

Student Travel Support, Connecticut Sports Analytics Symposium	2024, 2025, 2026
Sports Analytics Student Award, Carnegie Mellon Statistics & Data Science Funded by the Richardson Matano Family	2025
NFL Big Data Bowl Semifinalist	2025
NFL Big Data Bowl Finalist	2023, 2024
ASA Section on Statistics in Sports Student Funding	2024
Graduate Student Assembly/Provost Conference Funds, Carnegie Mellon University	2024
Reproducible Research Competition Finalist, Carnegie Mellon Sports Analytics Conference	2022
UTSPAN Scholarship, Cascadia Symposium on Statistics in Sports	2022
Reproducible Research Competition Winner, Carnegie Mellon Sports Analytics Conference	2021
Student Poster Award Winner, UConn Sports Analytics Symposium	2021
Merit Award, MS in Applied Statistics Program, Loyola University Chicago	2020 – 2021
Poster Competition Co-Winner, NextGen: Data Science Day, New England Statistical Society	2020

Summa Cum Laude , Wittenberg University	2020
Departmental Honors , Wittenberg Mathematics and Computer Science	2020
Phi Beta Kappa , Omicron of Ohio Chapter, Wittenberg University	2020
Pi Mu Epsilon , Ohio Upsilon Chapter, Wittenberg University	2020
Charles and Elsie Little Mathematics Award , Wittenberg Mathematics and Computer Science	2020
Awarded to the most outstanding senior majoring in mathematics and data science	
Mortar Board , Wittenberg University	2020
Pick and Pen , Wittenberg University	2019
Mathematics 202 Award , Wittenberg Mathematics and Computer Science	2018
Textbook Award , Wittenberg Mathematics and Computer Science	2018
Phi Eta Sigma , Wittenberg University	2018
Dean's List , Wittenberg University	All semesters; 2017 – 2020

SERVICES

Organizer

Carnegie Mellon Sports Analytics Center (CMSAC) Research Lab 2023 – present

Program Committee

UConn/Connecticut Sports Analytics Symposium 2024, 2025, 2026

Reviewer

Journal of Quantitative Analysis in Sports, Machine Learning, The American Statistician, WIREs Computational Statistics

Judge

CMSAC Reproducible Research Competition 2024, 2025

CSAS Student Poster Award 2024, 2025

CSAS Data Challenge 2024, 2025

Mentoring

Women in Data Science (WiDS) Chicago Datathon 2022

Statistical Consulting

Carnegie Mellon University Athletics 2024

Loyola University Chicago 2021 – 2022

(Clients: Stritch School of Medicine; Department of Biology; City of Milwaukee Assessor's Office)

MEDIA

Coverage & Mentions

NBC Sports, Carnegie Mellon University, Knowable Magazine, Amstat News, ABC News (Australia)

Podcast & Show Appearances

Open Source Sports Podcast, Viz Buzz (data science game show; multiple seasons)

SKILLS

Programming

R (expert); Python, SQL, SAS (experienced)

Tools

git/GitHub, Linux/bash (experienced)

Typesetting

markdown, $\text{\LaTeX}$ (proficient)

Bilingual

Vietnamese (native language) and English (fluent)

Last updated July 6, 2026